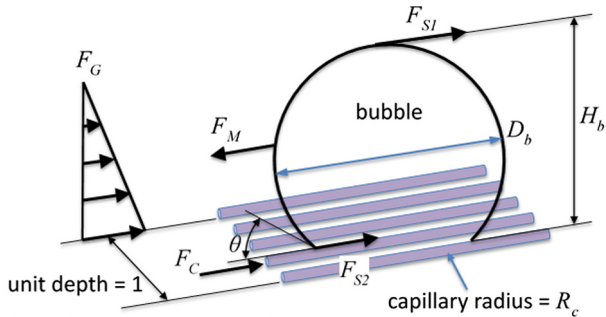




Number of tubes per unit depth = N

$$F_C = N\pi R_c^2 P$$

The number of tubes is directly proportional to the surface area

$$N \propto \frac{1}{S_m}$$

Using Equation (11) and the capillary force

$$F_C = 2\pi C \left(\frac{1}{S_m} \right)$$

In Equation (2), substituting Equation (11) results in

(11) results in

Fig. 12. Small capillary tubes beneath the bubble for modeling capillary wicking due to surface roughness.